

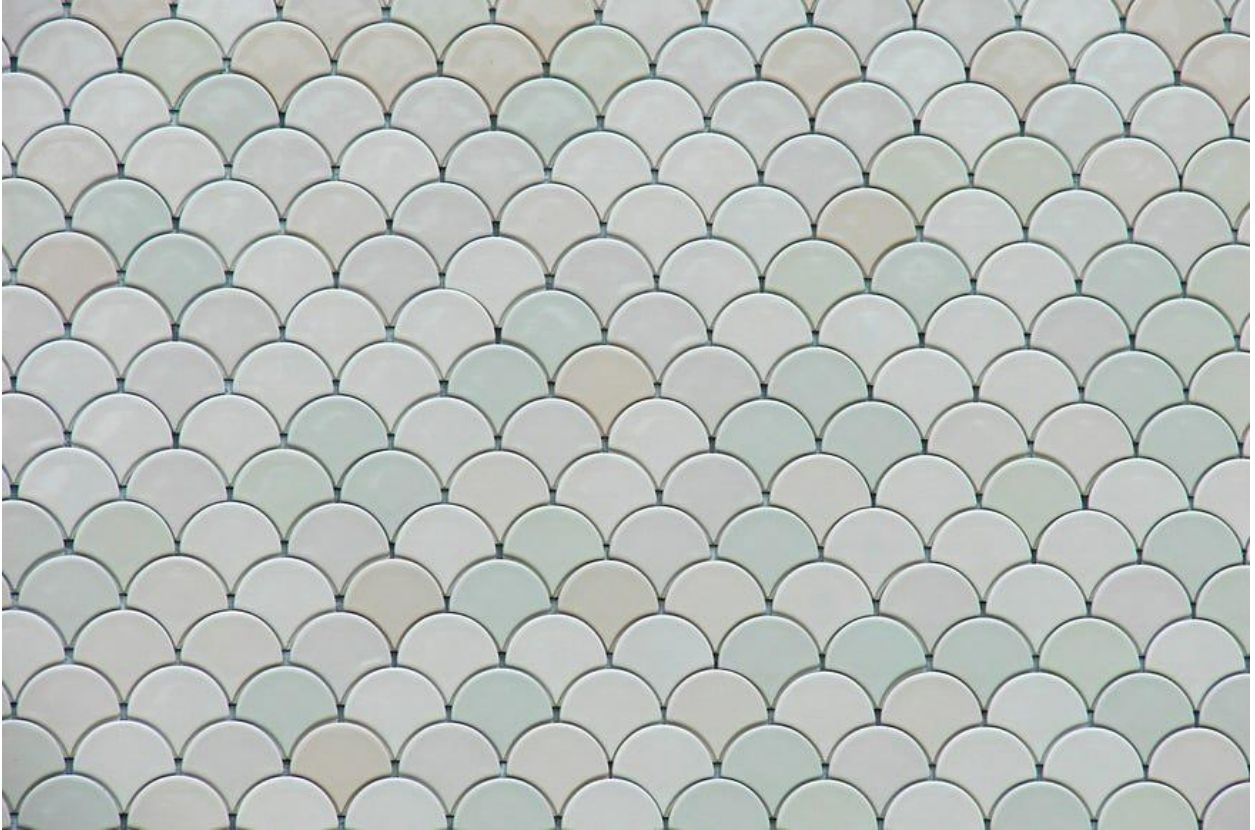


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Pattern-Seeking Creatures

How Our Brains Trick Us



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We are hard-wired to look for relationships and patterns in our environment. This is a huge evolutionary advantage and has enabled humans to adapt to a massive variety of regions and effectively exploit our surroundings. We can use this ability to find new sources of food, build new tools and shelters, and learn new things. Pattern seeking is truly our best skill.

Like all traits, there are downsides. Evolution inevitably results in missteps along the way, and our tendency to find patterns can misfire. Our brains' constant attempts to make sense of the world around us will frequently make mistakes. A common example is our tendency to find human faces in the most mundane objects, like the moon! Another classic pastime is to look for familiar shapes in clouds as they pass overhead, which is only possible because of how our brains work.

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This cross-section of a nematode worm has a creepy face in it! ([Source](#))

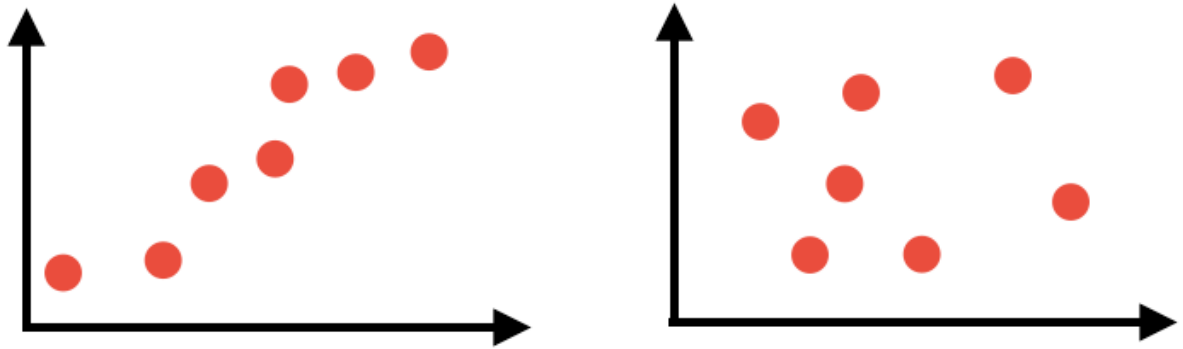
While our disposition for finding faces when they don't exist is fairly harmless, pattern recognition can sometimes have serious consequences. Many logical fallacies result from this tendency misfiring. The clearest example of this is the confusion between correlation and causation. This fallacy is at the core of many areas of pseudoscience.

These two words are a little tricky, so let's go over them again.

Correlation just means that two things are associated with each other. If it's cloudy outside, it's also more likely to rain. The taller someone is, the more they probably weigh. If you forget to drink your coffee in the morning, you're more likely to be tired at work. Someone who is born in the United States is not very likely to be fluent in Hungarian (I tried, it's a strange language!).

Notice that all of these examples are full of words such as “probably” and “likely.” Correlation does not mean two things will always happen together, just that they do so more often than not. Even if one event is associated with another event becoming less likely to occur, they would still be correlated. This is in contrast to two things being **independent**.

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These two abstract events are correlated (left) and independent (right). While the left plot shows a positive correlation, events can also be correlated negatively.

The second concept, **causation**, is a little trickier. There are countless definitions of what it means for something to cause something else. In this article, I'm going to avoid getting into the philosophical discussions of causation (if you want this, I have a link at the end of the article). You probably have an intuitive sense of what it means for something to cause something else, and that will work for the purposes of this article.

Many problems in science are dedicated to figuring out what causes what. If a person recovers from an illness, what made them get better? What causes an El Niño event to occur in the

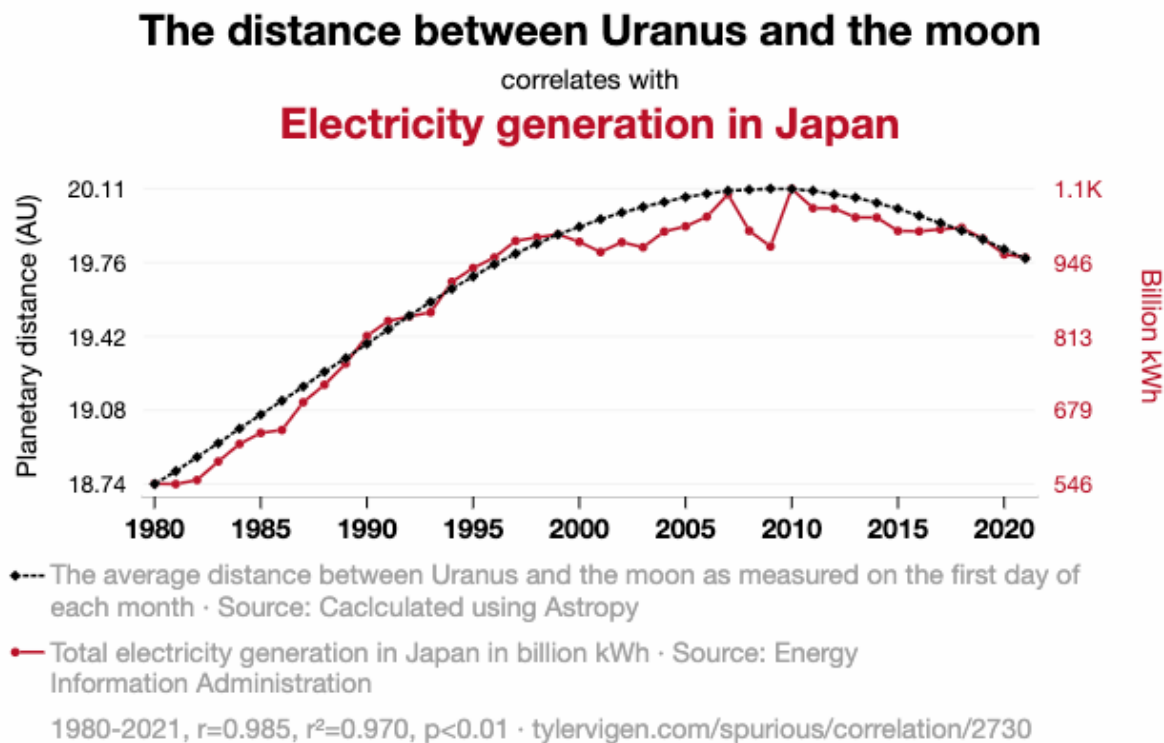
Pacific Ocean? What causes some people to be left-handed? Does smoking cause lung cancer?

Imagine seeing the following headline:

Months with higher ice cream consumption also have greater deaths by drowning! Avoid eating ice cream to lower your risk of drowning!

This report makes the classic mistake of confusing correlation with causation. It is true that drowning deaths and eating ice cream are *correlated*. But as the saying goes, **correlation does not imply causation**. Just because two things are correlated does not tell you that one causes the other. Here, there is a third variable, temperature, that is a factor. Months that are hotter both cause people to swim more and to eat more ice cream. As a result, these two quantities are correlated without any causal factor.

There is a website that compiles thousands of data sets on all sorts of topics. It then searches through the data to find correlations, no matter how ridiculous the match is. See the example below.

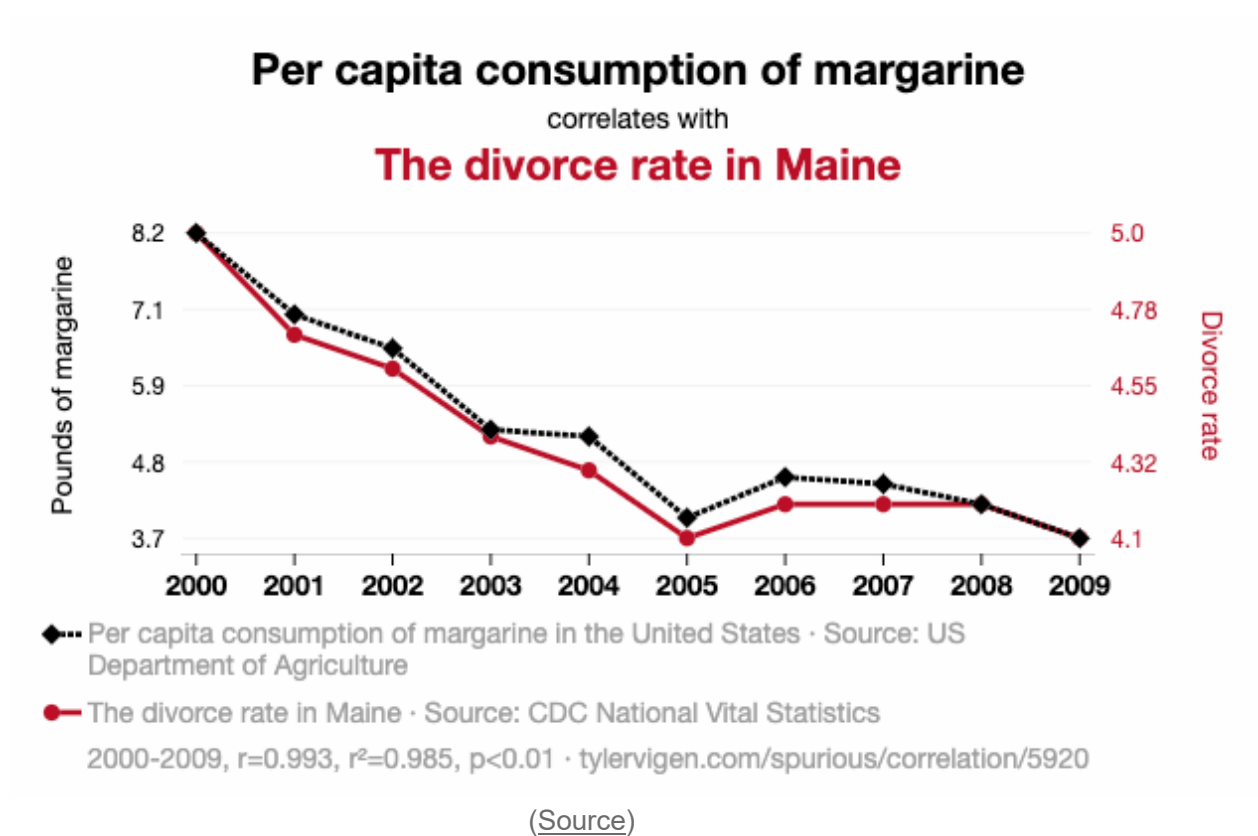


It looks like the distance between two celestial objects is driving power usage in Japan, or is it the other way around? ([Source](https://tylervigen.com/spurious/correlation/2730))

The correlation-causation fallacy would say that one of these variables is somehow causing the other! Even funnier, the site uses AI to generate a reason for how causality works and creates an entire academic article for each correlation.

This article is entirely AI-generated to “prove” that the distance between Uranus and the moon causes the amount of electricity generated in Japan to change ([Source](#))

Here’s another hilarious example.



While the two examples I’ve shown make it very obvious that these correlations are just a coincidence, it isn’t always that simple. Some events are difficult to disentangle. If you give someone a new drug to cure a problem they are having, did they get better because the drug actually did something?

It is well-documented that just doing anything to treat a patient will increase their risk of recovering, regardless of if the treatment actually was a real drug or not. Even after taking a useless sugar pill, patients often report improved symptoms from many illnesses. This is called the **placebo effect**. To make sure that a drug is actually doing something, there are steps that a scientists can take. It is common to randomly give half of your patients the real drug being tested, and the other half a “fake” drug that has no impacts. If the real treatment group shows an improvement better than the fake treatment group, we can be more confident that the drug is actually doing something.

Statistics is very good at identifying whether two variables are correlated or not. However, causality is more different to verify. Austin Bradford Hill, an English epidemiologist, was the first to introduce modern randomized trials like the one described above. He is most famous for discovering the link between smoking and lung cancer using these trials. These results were

highly controversial at the time of publication, but since are considered accepted facts.

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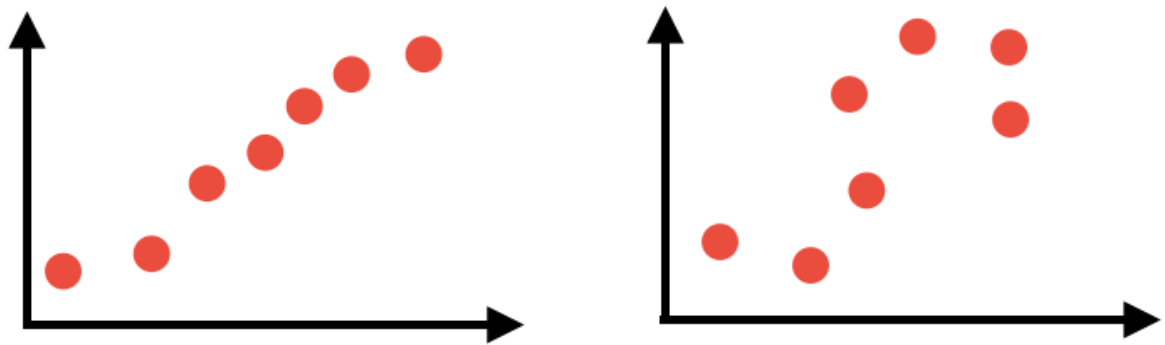


Austin Bradford Hill ([Source](#))

Hill outlined nine key factors that are important in establishing causality. While this list is applied for biological studies, these criteria are good for a variety of situations. Hill notes that it is rare to be able to find all nine of these in a single relationship, but the more you can find the better.

The two most important traits are **strength** and **consistency**. The more tightly associated two variables are, the better. A relationship should also have reproducibility, meaning it can be found in a variety of settings and samples. While these two factors are important, a relationship that is only strong and consistent may not indicate causality. See the Uranus-Moon distance and electricity example above, which satisfies both of these criteria but is clearly not causal.

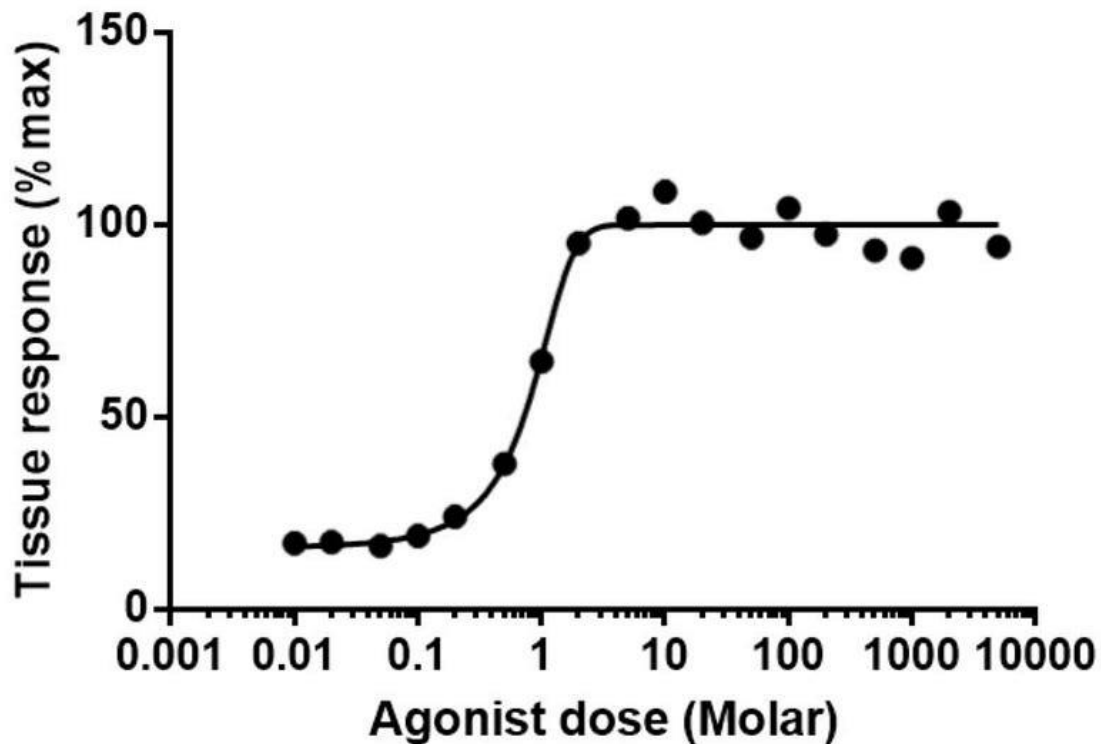
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While both are associated, the relationship on the left is much stronger

Similar to strength, Hill also suggests that a greater dose of input should lead to a stronger response which is also known as a **dose–response relationship**. In his original study, Hill found that the more frequently a person smoked, the greater their risk of lung cancer was. This criteria can be more variable depending on the mechanism you are investigating, and is sometimes an inverse relationship. Again, this trait still does not mean we are observing causation.

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The dose that was input (x-axis) and the observed response (y-axis) ([Source](#))

Hill's next two requirements make a causal relationship much more likely. You should establish how **specific** is the effect is. The narrower you can make the response and input variables and still have an association, the more likely you are observing a causal relationship. Additionally, Hill says the two events must be **consistent in time**. The response must occur after the input. While this might seem obvious, it rules out a lot of misleading correlations.

I'm going to group the next four traits together, because all are difficult to find but are strongly tied to proving a casual relationship. The most obvious way to prove causality is by running an **experiment**. This is like the randomized placebo trial I described earlier. Experiments where the input is randomly assigned to subjects are the gold standard for establishing causality. They effectively remove the possibility of a sneaky third variable messing up the results by driving the association. There are plenty of ways to make experimental results even stronger, like not telling patients if they are getting the treatment or not.

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However, experiments are expensive and can also be unethical. Imagine if Hill had randomly selected people to smoke 20 cigarettes a day, then observed how many got lung cancer? That would be a very questionable study and likely never get funded.

Luckily, there are three other tools we can use if an experiment isn't possible. The first is **plausibility**. Are there mechanisms that could explain why a causal relationship exists? This usually

requires a deep understanding of the underlying theory to do correctly, but is a valid way to theorize that one events causes another.

The other two are similar, so I will group them together.

Coherence and **analogy** are ways to make the case for a causal relationship. Does your proposed mechanism fit within a variety of other results that are already established? Then it is coherent. Does your relationship have similarities to any other associations? Then it is analagous.

These nine factors were laid out by Hill in a landmark paper The Environment and Disease: Association or Causation? (linked a the bottom). They have since been revised significantly: notice that several of them could be easily group together. It is difficult to overstate how huge of an impact this paper had on the medical field; it has been cited over 13,000 times!

Overall, it is much more difficult to show that one variable causes another. This is a constantly evolving area of study, and we are getting better and better at our ability to do science! Knowing the difference between correlation and causation is so important to your ability to think critically about what we see online and on the news.

Going Further

I hope you learned something! The intersection of mathematics and logical fallacies is fascinating. Significant work has been done over the years to improve our ability to analyze relationships within datasets. If you want to learn more about this topic, I've included some links below for you to check out.

[How Not to Be Wrong: The Power of Mathematical Thinking](#) is a fantastic book about fallacies in statistics and how to correct for them

There are tons of free online stats resources out there. I use [Introduction to Modern Statistics](#) in my class, which is free and open source

The Stanford Encyclopedia of Philosophy has some fascinating articles on what causation really means; [I've linked one of them here.](#)

Hill's original paper is available at [this link](#) and is free to download.

I also have some similar articles to this one that you may be interested in. Check them out!